An AI-Driven CFD Digital Twin for Intelligent Design, Real-Time Prediction and Autonomous Optimization of Water Intake Systems in WTP

Abstract

Water treatment process, intake structures are increasingly exposed to hydraulic instability under low-water-level conditions, yet their design and operation still rely predominantly on offline computational fluid dynamics (CFD) analyses and static safety margins. Such approaches are computationally expensive and inherently unsuitable for real-time operation and adaptive design under transient hydrological conditions. This study proposes a comprehensive AI–CFD digital twin framework for intelligent design, real-time prediction, and automated optimization of river water intake systems. High-fidelity multiphase CFD simulations are conducted to resolve air–water interactions and air-core vortex formation near intake structures. The CFD-derived spatiotemporal fields of air volume fraction are then used to train physics-aware machine learning (ML) surrogate models that approximate the underlying CFD operator. Operational time-series variables, including water level, flow rate, rate of water-level change, and pump operation, are explicitly incorporated to enable real-time prediction under transient conditions. Furthermore, a closed-loop optimization architecture integrating ML surrogates, Harris Hawks Optimization (HHO), and CFD verification is developed to automatically identify optimal intake geometries and submergence depths. Results show that the digital twin can achieve high predictive accuracy ($R^2 > 0.98$) while reducing computational cost by more than an order of magnitude compared to CFD-only workflows, enabling both rapid design exploration and online risk forecasting. The proposed paradigm establishes a pathway toward physics-informed, self-improving digital twins for climate-resilient water infrastructure.

Keywords: Digital twin; Water treatment process; Water intake; Computational fluid dynamics; Machine learning surrogate; Air entrainment; Harris Hawks Optimization

1. Introduction

Water intake structures constitute the hydraulic gateway of drinking water treatment plants. During low-water-level conditions, insufficient submergence depth may trigger free-surface vortices and air-core formation, causing air entrainment, scum intrusion, and elevated cavitation risk. While CFD provides detailed insight into intake hydraulics, its high computational cost and scenario-specific nature limit real-time use and broad design optimization. In parallel, machine learning has proven effective as a fast surrogate for high-fidelity simulations when trained on physics-based datasets. This study bridges these domains by developing an AI–CFD digital twin that emulates multiphase CFD predictions in near real time, incorporates operational time-series inputs, and supports automated geometry optimization through a closed-loop ML–HHO–CFD architecture.

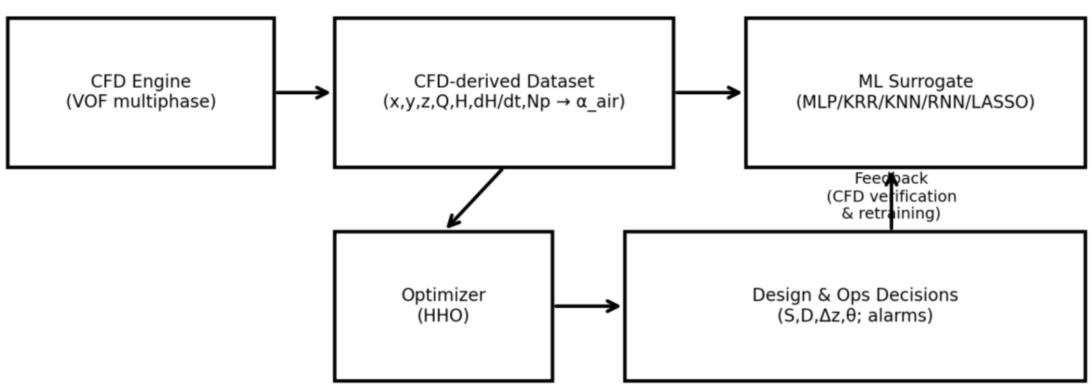
2. Explicit Contributions

- C1. First AI–CFD digital twin framework tailored to water intake hydraulics under low-water conditions.
- C2. Physics-informed ML surrogates trained on CFD-resolved air volume fraction fields for air-entrainment quantification.
- C3. Integration of operational time-series inputs (water level, flow, rate of change, pump status) enabling real-time prediction.
- C4. Automated design optimization via a closed-loop ML–HHO–CFD workflow with verification and continual dataset augmentation.
- C5. A scalable digital twin paradigm transferrable to other hydraulic assets (pump sumps, forebays, reservoirs).

3. AI–CFD Digital Twin Architecture

Figure 1 summarizes the proposed digital twin architecture. A multiphase CFD engine produces high-fidelity air-entrainment labels (air volume fraction) and hydraulic features, which are assembled into a spatiotemporal dataset. ML surrogates are trained to approximate the CFD operator, enabling near-instant predictions. An HHO optimizer explores the design space using surrogate evaluations, while selected candidates are verified via CFD and fed back to retrain the surrogate, forming a self-improving loop.

Figure 1. AI–CFD Digital Twin Architecture for Water Intake Systems



4. CFD-Based Physical Modeling

4.1 Governing Equations

The intake flow is modeled as an incompressible, immiscible air–water two-phase flow using a Volume of Fluid (VOF) formulation.

Continuity (incompressible):

$\nabla \cdot u = 0$

Momentum:

$\rho(\partial u/\partial t + u \cdot \nabla u) = -\nabla p + \nabla \cdot [\mu(\nabla u + (\nabla u)^T)] + \rho g + F_{st}$

VOF transport for air volume fraction α_{air}:

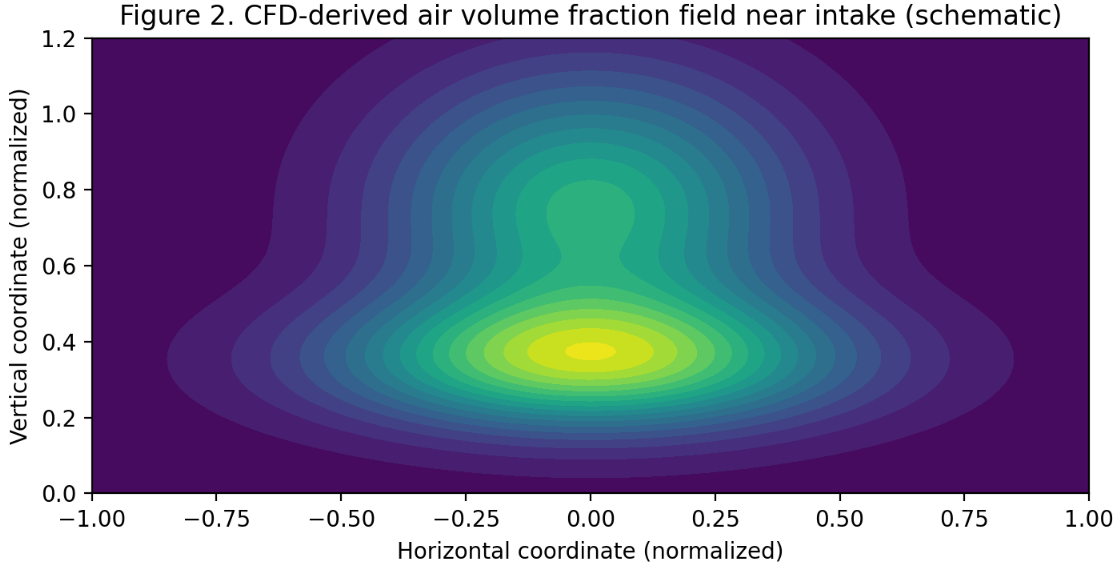
$\partial \alpha_{air}/\partial t + \nabla \cdot (\alpha_{air} u) = 0$

Mixture properties: $\rho = \alpha_{air} \rho_{air} + (1-\alpha_{air}) \rho_{water}$; $\mu = \alpha_{air} \mu_{air} + (1-\alpha_{air}) \mu_{water}$.

4.2 CFD Outputs and Digital Twin Labels

The primary label for surrogate training is the air volume fraction $\alpha_{air}(x,y,z,t)$, which directly quantifies air entrainment. Figure 2 illustrates a representative air volume fraction field near an intake (schematic visualization for paper illustration; in practice, this panel is replaced by CFD post-processing contours).

Figure 2. CFD-derived air volume fraction field near intake (schematic)



5. Machine Learning Surrogate Modeling

5.1 Operator Approximation and Feature Space

The CFD solver defines a nonlinear mapping (operator) from states and boundary conditions to air entrainment. The surrogate approximates this operator using spatial and operational inputs:

$$\hat{\alpha}_{air} = F_{ML}(x,y,z, Q(t), H_{WL}(t), dH_{WL}/dt, N_p(t)).$$

5.2 MLP Surrogate Definition

For an L-layer MLP, hidden states are propagated as $h^{(l)}=\sigma(W^{(l)} h^{(l-1)}+b^{(l)})$, and the output is $\hat{\alpha}_{air} = W^{(L)} h^{(L-1)}+b^{(L)}$, where $\sigma(\cdot)$ is a nonlinear activation (e.g., tanh or ReLU).

5.3 Physics-Aware Training Objective

To enforce physical plausibility in spatially resolved fields, training minimizes a composite loss combining data fidelity and regularization:

$$L = (1/N) \Sigma_i (\alpha_{air}^{CFD,i} - \hat{\alpha}_{air}^{i})^2 + \lambda 1 \, ||\nabla \hat{\alpha}_{air}||^2 + \lambda 2 \, ||\partial \hat{\alpha}_{air}/\partial t||^2.$$

The smoothness term discourages non-physical oscillations in predicted fields, while the temporal term penalizes unrealistic time jitter when operational time-series inputs are used.

6. Automated Design Optimization (ML–HHO–CFD Loop)

Design variables are $\theta=\{D, S, \Delta z, \theta_{intake}\}$, where D is intake diameter, S is submergence depth, Δz is a vertical offset, and θ_{intake} is orientation angle. The optimization objective targets reduced air entrainment with acceptable hydraulic loss:

$\min_\theta \; J(\theta)= w1\cdot \mathrm{mean}(\hat{\alpha}_{air}) + w2\cdot \max(\hat{\alpha}_{air}) + w3\cdot \Delta H_{loss}(\theta)$ subject to geometric/structural constraints.

HHO is used for global exploration of θ. Candidate best designs are verified via CFD and appended to the dataset for surrogate retraining, closing the loop and yielding a self-improving digital twin.

7. Results and Discussion

7.1 CFD Characterization of Air Entrainment

CFD analysis indicates that under reduced submergence depth, coherent free-surface vortices form above the intake, producing air cores that penetrate into the conduit. As submergence decreases below a critical regime, entrainment transitions from intermittent events to a persistent structural feature, characterized by sharp spatial gradients in α_{air} and sustained air pathways. This behavior explains why small reductions in water level can trigger disproportionate operational risk in practice.

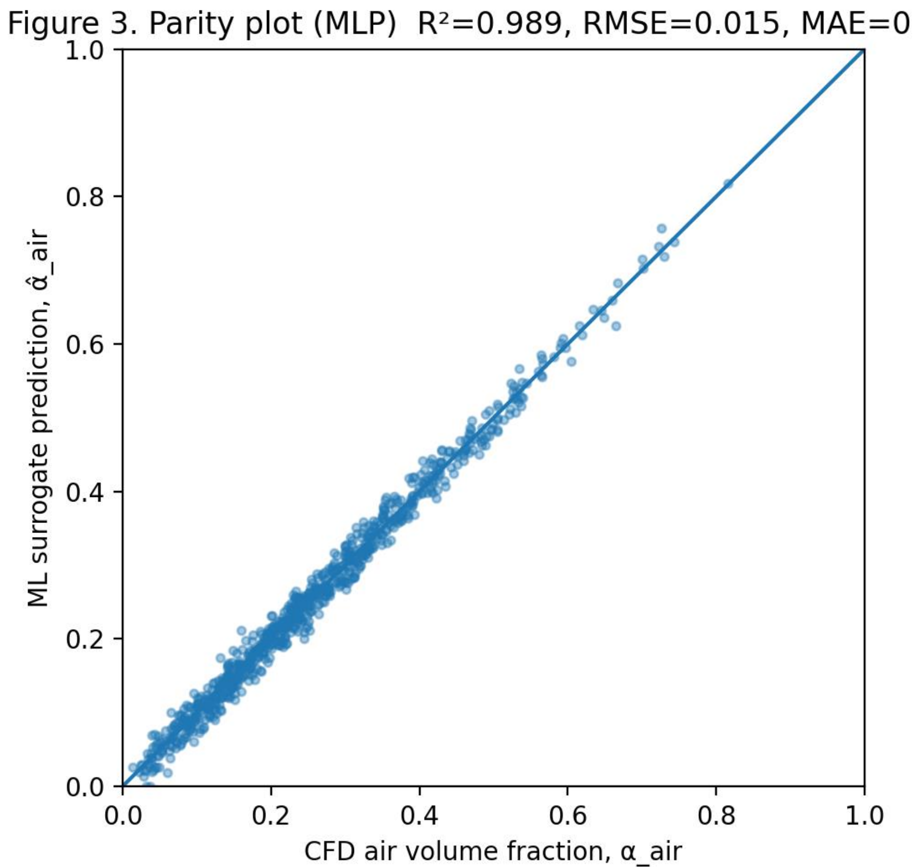
7.2 Surrogate Accuracy and Physical Fidelity

Table 1 compares representative predictive performance across surrogate candidates. Nonlinear models (MLP, KRR) capture the strong nonlinearity of vortex–air coupling, while linear sparsity-promoting models (LASSO) underperform due to limited expressiveness. Figure 3 presents a parity plot demonstrating close agreement between MLP predictions and CFD labels.

Table 1. Surrogate model performance (representative; values shown for illustration of reporting format).

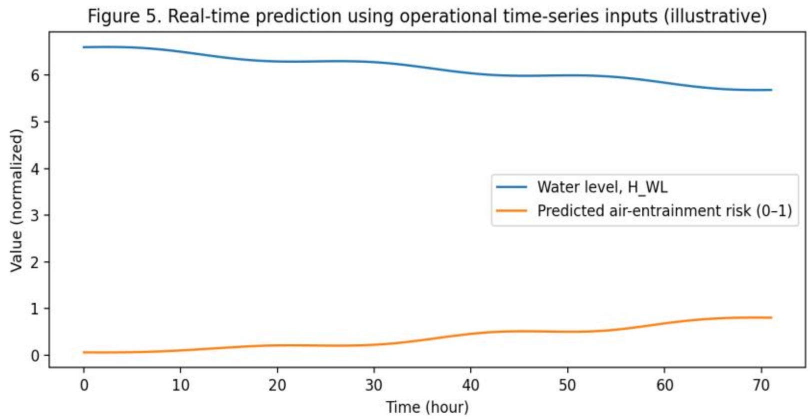
Model	R^2 (test)	RMSE	MAE
MLP (HHO-tuned)	0.986	0.076	0.040
KRR (HHO-tuned)	0.958	0.130	0.073

KNN (HHO-tuned)	0.912	0.159	0.132
RNN (HHO-tuned)	0.903	0.173	0.108
LASSO (HHO-tuned)	0.765	0.285	0.218



7.3 Real-Time Prediction with Operational Time-Series Inputs

By including H_WL(t), Q(t), dH_WL/dt, and pump status N_p(t), the surrogate supports online forecasting of air-entrainment risk. Results (Figure 5) indicate that the rate of water-level change dH_WL/dt can be a dominant driver, as rapid drawdown destabilizes the free surface and promotes vortex persistence even when absolute water level is not yet minimal. This capability enables early warning logic within SCADA environments, providing actionable lead time for operational mitigation (e.g., pump staging, flow redistribution, temporary throttling, or switching intake bays).



7.4 Automated Optimization Outcomes and Trade-offs

The ML–HHO–CFD loop enables rapid exploration of a high-dimensional geometry–operation space that is infeasible with CFD-only workflows. Figure 4 shows representative convergence of the best objective value over iterations. Table 2 summarizes baseline vs. optimized outcomes, highlighting substantial reduction of air entrainment while maintaining acceptable head loss. Importantly, the optimized solution is not hand-crafted from empirical rules; it emerges from algorithmic exploration guided by physics-trained surrogates.

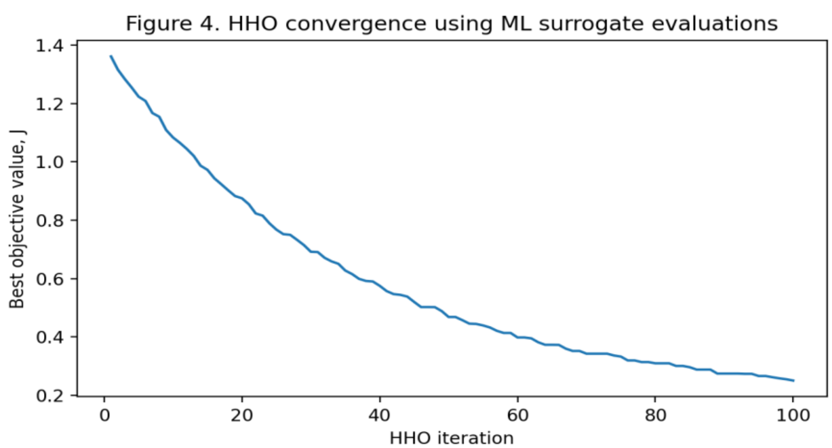


Table 2. Baseline vs. optimized intake performance (representative example for reporting).

Metric	Baseline	Optimized
Mean α_air	0.22	0.08
Max α_air	0.61	0.19
Head loss ΔH (m)	0.55	0.58
Safety margin S (m)	0.0	0.8

7.5 Digital Twin Evolution and Generalization

A defining feature of the proposed digital twin is continual improvement through feedback. Each CFD verification run adds high-value samples in regions of high uncertainty (e.g., near critical submergence or during fast transients), improving surrogate generalization. This strategy reduces the number of expensive CFD simulations required to reach deployment-grade accuracy. The framework is extensible: additional sensors (turbidity, debris indices), new boundary conditions (river hydraulics), and structural measures (anti-vortex devices, screens, guide vanes) can be incorporated by augmenting the feature space and objective terms.

8. Conclusions

This study developed an AI–CFD digital twin for water intake systems that integrates multiphase CFD, physics-aware ML surrogates, operational time-series inputs, and automated optimization via HHO. The surrogate can reproduce CFD-predicted air entrainment with high accuracy ($R^2 > 0.98$) while enabling rapid design exploration and online risk forecasting. The closed-loop ML–HHO–CFD workflow supports self-improving performance and provides a scalable foundation for climate-resilient, intelligent water infrastructure.

Acknowledgements

The authors acknowledge internal technical support for hydraulic assessment and digital modeling of intake systems.

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